Marginal Probability

(Q1) A bag contain 6 red balls, 5 green balls, 4 blue balls.
one ball is drawn at random.
find the Probability that ball drawn is

(a) green

$$P = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

$P(\text{green}) = \frac{5}{15} = \frac{1}{3}$

(b) not blue

$P(\text{not blue}) = \frac{11}{15} = \frac{11}{15}$

(Q2) A bag contain 5 red balls and 5 blue balls one ball is drawn at random
find the Probability that balls drawn

(a) red

$$P = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

$P(\text{red}) = \frac{5}{10} = \frac{1}{2}$

$P(\text{red}) = \frac{1}{2}$

(2) blue:

$P(\text{blue}) = \frac{5}{10} = \frac{1}{2}$

$P(\text{blue}) = \frac{1}{2}$

Q3) A box contain 8 good bulbs and 2 defective bulbs. one bulb is selected at random.

Find Probability that the selected bulb is

(a) good

$$P(\text{good}) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$$

$$= \frac{8}{10} = \frac{4}{5}$$

$P(\text{good}) = \frac{4}{5}$

(b) defective:-

$$P(\text{defective}) = \frac{2}{10} = \frac{1}{5}$$

Additive Probability:

Q4. A Card is drawn at random from standard deck of 52 playing cards

find the Probabity that the card drawn

(a) a diamond.

number of diamonds = 13

$$P(\text{diamond}) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$$

$$\Rightarrow \frac{13}{52}$$

$$\Rightarrow \frac{1}{4}$$

(b) a face card.

face cards = 12.

$$P(\text{face cards}) = \frac{12}{52} = \frac{3}{13}$$

(c) Diamond (or) face card.

union rule $= P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A) = 13$ $P(B) = 12$ $P(A \cap B) = 3$

$$P = \frac{13}{52} + \frac{12}{52} - \frac{3}{52}$$

$$P = \frac{13+12-3}{52} = \frac{22}{52} = \frac{11}{26}$$

(Q5) A fair die is rolled once.

(a) Prime number

Prime numbers in die = 2, 3, 5.

$P(\text{Prime numbers}) = \frac{\text{number of favorable outcomes}}{\text{Total number of outcomes}}$

$P(\text{Prime numbers}) = \frac{3}{6} = \frac{1}{2}$

(b) number less than 3

number less than 3 = 1, 2.

$P(\text{number less than } 3) = \frac{2}{6} = \frac{1}{3}$

(c) a prime number (or) number less than 3

Prime number = 2, 3, 5

number < 3 = 1, 2.

union = 1, 2, 3, 5

$P = \frac{4}{6} = \frac{2}{3}$.

(Q6) In a group of 50 students, 28 like Python, 22 like SQL and 10 like both Python and SQL

(a) Likes Python (or) SQL.

(Q6)

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \frac{28}{50} + \frac{22}{50} - \frac{10}{50}$

$\Rightarrow \frac{40}{50}$

$\Rightarrow \frac{4}{5}$.

(b) Probability that a student Likes neither

Students who Like atleastone = 40

total Student = 50

Students Like neither = $50 - 40 = 10$

$P(\text{neither}) = \frac{10}{50}$

$\Rightarrow \frac{1}{5}$

(Q7) In a survey of 200 People, 90 People use Instagram, 70 uses twitter uses both.

(a) the Probability that a random selected Person uses Instagram

$P(I) = \frac{90}{200}$

$\Rightarrow \frac{9}{20}$.

(b) the Probability that Person uses twitter.

$P(T) = \frac{70}{200}$

$\Rightarrow \frac{7}{20}$

(c) Probability that a Person uses Instagram or twitter.

$P(I \cup T) = P(I) + P(T) - P(I \cap T)$

$P(I \cup T) = \frac{90}{200} + \frac{70}{200} - \frac{40}{200} = \frac{120}{200} =$

$P(I \cup T) = \frac{3}{5}$

(Q8) A class has 30 students out of them 18 like mathmatics, 12 likes statistics and 6 likes both.

(a) Likes mathmatics (or) statistics

$P(m \cup s) = P(m) + P(s) - P(m \cap s)$

$\Rightarrow \frac{18}{30} + \frac{12}{30} - \frac{6}{30} = \frac{24}{30}$

$P(m \cup s) \Rightarrow \frac{4}{5}$

(b) Probability that a students like only mathmatics

maths = 18

both = 6

$18 - 6 = 12$

$P = \frac{12}{30} = \frac{2}{5}$

(c) Likes neither maths nor stats.

~~maths~~ both = 6

$P(\text{neither}(s)) = \frac{6}{30}$

$P = \frac{1}{5}$

Joint Probability

(Q9) Two fair coins are tossed simultaneously.

(a) exactly one head.

Possible outcomes: {HH, HT, TH, TT}

exactly one head: HT, TH.

$$P(\text{exactly one head}) = \frac{\text{no. of favourable outcomes}}{\text{total no. of outcomes}}$$

$$\Rightarrow \frac{2}{4}$$

$$\Rightarrow \frac{1}{2}.$$

(b) Probability of atleast one head.

Possible atleast one head: HH, HT, TH.

$$P(\text{atleast one head}) = \frac{3}{4} = \frac{3}{4}.$$

(Q10) A coin is tossed and die is rolled simultaneously

(a) a tail and number greater than 4.

number greater than 4 on die: 5, 6.

favorable outcomes

- T5
- T6.

$$P(T \text{ and } > 4) = \frac{2}{12}$$

$$\Rightarrow \frac{1}{6}$$

(b) Head and an even number.

even number on die = 2, 4, 6

favourable outcomes = H2, H4, H6.

$$P(\text{Head and even number}) = \frac{3}{12}.$$

$$\Rightarrow \frac{1}{4}.$$

(Q11) Two fair dice are rolled together.

(a) Sum of numbers obtained is 8.

Total = 2.

$P(A \cap B) = \frac{2}{52}$

$\Rightarrow \frac{1}{26}$

(ii) $P(A \cup B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \frac{26}{52} + \frac{4}{52} - \frac{2}{52}$

$\Rightarrow \frac{28}{52} = \frac{7}{13}$

$P(A \cup B) = \frac{7}{13}$

Q13) Conditional Probability

In a class of 40 students, 24 student passed maths, 20 students passed science and 8 passed both.

(a) the probability that student passed maths give that he/she passed science.

$P(M|S) = \frac{8}{20}$

$\Rightarrow \frac{2}{5}$

(b) Probability that a student passed science given that the student passed maths

$P(S|M) = \frac{P(M \cap S)}{P(M)}$

$= \frac{8}{24}$

$P(S|M) = \frac{1}{3}$

Q14) A card is drawn random from standard deck 52 cards given that the drawn is black find the probability that is a spade.

Total = 2.

$P(A \cap B) = \frac{2}{52}$

$\Rightarrow \frac{1}{26}$

(ii) $P(A \cup B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \frac{26}{52} + \frac{4}{52} - \frac{2}{52}$

$\Rightarrow \frac{28}{52} = \frac{7}{13}$

$P(A \cup B) = \frac{7}{13}$

Q13) Conditional Probability

In a class of 40 students, 24 student passed maths, 20 students passed science and 8 passed both.

(a) the probability that student passed maths give that he/she passed science.

$P(M|S) = \frac{8}{20}$

$\Rightarrow \frac{2}{5}$

(b) Probability that a student passed science given that the student passed maths

$P(S|M) = \frac{P(M \cap S)}{P(M)}$

$= \frac{8}{24}$

$P(S|M) = \frac{1}{3}$

Q14) A card is drawn random from standard deck 52 cards given that the drawn is black find the probability that is a spade.

When n and s = 26
spade s = 13.

$S \cap G = 13$

$P(S|G) = \frac{13}{26}$

$= 1/2$

(Q15) A jar contain 4 red balls and 6 white balls. two balls are drawn one after the other without replacement. find the probability the second ball drawn is white, given that the first ball drawn was white.

red = 4
white = 6
total = 10

first ball is white, one white is removed

white = 5
red = 4
total = 9.

Probability the second ball is white.

$P(W_2|W_1) = \frac{\text{remaing white balls}}{\text{remaing total balls}}$

$= \frac{5}{9}$

Descriptive statistics

(Q16) A class 10 students. marks are.
35, 40, 45, 50, 50, 55, 60, 70, 75, 80
(a) median.

n is even the median is the mean of 5th and 6th.

$\text{median} = \frac{50+55}{2} = 52.5$

(b) IQR.

lower half:
35, 40, 45, 50, 55

$Q_1 = 45$

upper half:
55, 60, 70, 75, 80

$Q_3 = 70$

$IQR = Q_3 - Q_1 = 70 - 45 = 25$

(c) check for outliers.

outliers are.

lower $= Q_1 - 1.5(IQR)$

$45 - 1.5(25)$

$\Rightarrow 45 - 37.5 = 7.5$

upper $= Q_3 + 1.5(IQR)$

$\Rightarrow 70 + 37.5 = 107.5$

outliers

$7.5 \leq x \leq 107.5$.

No outliers are exist

3.17. consider the following waiting times.

4.21, 5.55, 3.02, 5.13, 4.77, 2.34, 3.54, 3.29, 4.50, 6.10, 0.38, 5.
6.46, 6.19, 3.79

(a) find the five number summary.

1 minimum:-

minimum = 0.38.

(2) median:

Sorted values.

0.38, 2.34, 3.02, 3.20, 3.54, 3.79, 4.21, 4.50, 4.77, 5.
5.13, 5.55, 6.10, 6.19, 6.46

Ainter: $\frac{9+1}{2} = \frac{15+1}{2} = 8$

redian = 4.50

3. Q1

lower: 0.38, 2.74, 3.02, 3.20, 3.54, 3.74, 4.21

redian = 3.20

Q1 = 3.21

4. Q3.

upperhalf: 4.77, 5.21, 5.13, 5.55, 6.10, 6.19, 6.46

redian 5.55

(5) maximum:-

maximum = 6.46

Q(18) A retaological agency conducted 13 high altitude weather observation mission b/w 2020 and 2024.

9, 195, 241, 301, 216, 260, 7, 244, 192, 147, 10, 295, 142

mean:-

$$\bar{x} = \frac{\text{Sum of observation}}{\text{number of observations}}$$

$$\frac{9+195+241+301+216+260+7+244+192+147+10+295+142}{13}$$

$$= \frac{2254}{13}$$

$\bar{x} = 173.77$

Standard deviation:

$$S = \sqrt{\frac{\Sigma (x-\bar{x})^2}{n-1}}$$

$\bar{x} = 173.77$.

$$\sqrt{\frac{(9-173.77)^2+(195-173.77)^2+(241-173.77)^2+(361-173.77)^2 \ldots\ldots}{13-1}}$$

$$\Rightarrow \sqrt{\frac{166241.54}{12}}$$

$$\Rightarrow \sqrt{13853.46}$$

$$S \Rightarrow 117.70$$

(19) A sample of ten weights (kg) is given

30 32, 35 38, 40 42, 45, 48, 50, 53

Find:-

(a) Q_1, Q_2, Q_3.

(Q_1) first half:

30, 32, 35, 38, 40

median of the 5 = 35

$Q_1 = 35$

(Q_2) median of data $= \frac{40+42}{2} = 41$

(Q_3). 42, 45, 48, 50, 53 $\cdot$ $Q_3 = 48$

(b) IQR.

$IQR = Q_3 - Q_1 = 48 - 35 = 13$

(c) outliers using 1.5 × IQR.

$1.5 \times IQR = 1.5 \times 13 = 19.5$

lower bound:

$Q_1 - 1.5(IQR) = 35 - 19.5 = 15.5$

upper bound $Q_3 + 1.5(IQR) = 48 + 19.5 = 67.5$

outliers range $= 15.5 \leq x \leq 67.5$

all values are inside range 30 to 53.